Bio diesel

Bio diesel is a fuel obtained from vegetable oils such as soya bean oil, jatropha oil, corn oil, sunflower oil, cotton seed oil, rice-bran oil and rubber seed oil.



Hydrogen - The future fuel

Hydrogen could be the best alternative fuel in the future. It is a clean fuel as it gives out only water while burning. Moreover, it has the highest energy content. It does not pollute air.

Wind energy

Wind energy is obtained with the help of wind mills. When wind blows, they rotate the blades of the wind mills and electricity is produced by the dynamo. Wind mills are mostly located at Kayathar, Aralvaimozhi, Palladam and Kudimangalam in Tamil Nadu.



Figure 15.18 Wind Mill

Gobar Gas

Gobar gas is obtained by the fermentation of cow dung in the absence of air (anaerobic conditions). It mainly contains methane and a little ethane. It is widely used in rural areas for cooking and operating engines.

15.8 Solar Energy

Sun is the first and foremost energy source that makes life possible on the earth. Solar energy is the only viable fuel source of non depleting nature for sun provides a free and renewable source of energy. It is the renewable type of energy without endangering the environment. It is the potential source to replace the fossil fuel in order to meet the needs of the world. With the advancements in science and technology, solar energy has become more affordable, and it can overcome energy crisis. Solar energy is a clean energy. With the minimum efforts maximum energy can be harnessed using various equipments.

15.8.1 Applications of Solar Energy

Solar energy has wider applications in various fields.

- It is used in solar water heater.
- It is used in drying of agricultural and animal products.
- It is used in electric power generation.
- It is used in solar green houses.
- It is used in solar pumping and solar distillation. It is used for solar cooking and solar furnaces also.



Figure 15.19 Solar panel

Points to Remember

- Methane is the simplest hydrocarbon in which four hydrogen atoms are linked with one carbon atom.
- Natural gas is a naturally occurring hydrocarbon gas mixture consisting primarily of methane.
- CNG is the cheapest and cleanest fuel. Vehicles using this gas produce less carbon dioxide and hydrocarbon emission. It is less expensive than petrol and diesel.
- Producer gas is a gaseous mixture of carbon monoxide and nitrogen.

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